Kinds of collisions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate momentum conservation from kinetic-energy conservation.

SUCCESS CRITERIA - I can...

1. Identify the system and external-impulse assumption.
2. Recognize sticking as perfectly inelastic.
3. Use kinetic energy to distinguish elastic from inelastic.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “collision” and “elastic” to explain your model or result. Sentence starters are optional.

collision: an interaction during which objects exert forces on each other

elastic: a collision that conserves total kinetic energy

inelastic: a collision that transforms some kinetic energy to other forms

01 Warm-up: bring back an earlier idea

A positive triangular force pulse lasts 0.3 s and peaks at 50 N.
Find impulse and velocity change for a 0.6 kg object.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{total,\,i} = p_{total,\,f}$; $K = \frac{1}{2}mv^2$.
Negligible external impulse. Elastic collisions also have $K_{total,\,i} = K_{total,\,f}$; sticking transforms some kinetic energy.

Worked reasoning: A 1 kg cart at 4 m/s joins a 1 kg cart at rest. Initial momentum is 4 kg·m/s. Together, 2 kg move at 2 m/s; sticking is one possible collision outcome.

Your turn: Draw before and after snapshots with masses and velocities.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch before/after for a collision where two carts stick.
Label one common final velocity, not two different final velocities.

Sentence starter: Both … . They differ in … because … .

Before contact: add velocity arrows.



Cart A Cart B

Marzano check - circle: 1 2 3

04 Apply the physics

A 2 kg cart traveling at +3 m/s sticks to a stationary 1 kg cart on a low-friction track.
Draw and label the collision.
Find the common final velocity, kinetic energy before and after, and fraction of initial kinetic energy transformed.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A classmate says energy is destroyed in an inelastic collision. Rewrite the claim accurately.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A projectile launches horizontally at 6 m/s from height 45 m.
Use g=10 and no drag.
Find flight time, range and final vertical velocity.
Label separate horizontal and vertical motion.

Word help: horizontal = along a level line, usually the x direction

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Two equal masses collide elastically head-on, one initially at rest.
Give their final velocities relative to initial moving speed u. Organize the known quantities on the before-and-after model.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I notice … . This supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. For the same initial state, test final velocities (0, 4), (1, 3) and (2, 2) m/s for momentum conservation. Which pairs represent sticking? Organize the known quantities on the before-and-after model.

Sentence starter: I notice … . This supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3